◈ 1. Repository Setup

Initialize / Clone

```
git init
git clone <url>
git clone <url> <folder-name>
```

◈ 2. Basic Snapshot Workflow

Check status

```
git status
```

Stage changes

```
git add <file>
git add .
git add -A
```

Commit

```
git commit -m "message"
git commit -m "message"   # tracked files only
```

View changes

```
git diff
git diff --staged
```

◈ 3. Viewing History

```
git log
git log --oneline
git log --graph --oneline --all
git show <commit-id>
git blame <file>
```

◈ 4. Branching & Switching

```
git branch
git branch <branch-name>
git checkout <branch>
git checkout -b <branch>
git switch <branch>
git switch -c <branch>
git branch -d <branch>
```

◈ 5. Merging & Rebasing

Merge

```
git merge <branch>
```

Rebase

```
git rebase <branch>
git rebase -i <commit>
```

Abort operations

```
git merge --abort
git rebase --abort
```

◈ 6. Remote Repositories (e.g., GitHub)

```
git remote -v
git remote add origin <url>
git fetch
git pull
git pull origin main
git push
git push origin <branch>
git push -u origin <branch>
```

◈ 7. Undoing Changes (Very Important in Interviews)

Unstage file

```
git reset <file>
```

Undo commit (keep changes)

```
git reset --soft HEAD~1
```

Undo commit (discard changes)

```
git reset --hard HEAD~1
```

Revert commit (safe for shared history)

```
git revert <commit-id>
```

Discard local changes

```
git checkout -- <file>
git restore <file>
```

◈ 8. Stashing

```
git stash
git stash save "message"
git stash list
git stash apply
git stash pop
git stash drop
```

◈ 9. Tagging

```
git tag
git tag v1.0
git tag -a v1.0 -m "release"
git push origin --tags
```

◈ 10. File & Commit Inspection

```
git ls-files
git cat-file -p <hash>
git reflog
```

👉 git reflog is **frequently asked** (recover lost commits).

◈ 11. Cleaning & Maintenance

```
git clean -n
git clean -f
git gc
```

◈ 12. Advanced / Power Commands (High Interview Value)

Cherry-pick

```
git cherry-pick <commit>
```

Squash commits

```
git rebase -i HEAD~n
```

Amend last commit

```
git commit --amend
```

Bisect (debugging)

```
git bisect start
git bisect bad
git bisect good <commit>
```

◈ 13. Configuration

```
git config --global user.name "Your Name"
git config --global user.email "email@example.com"
git config --list
```

◈ 14. Alias (Productivity)

```
git config --global alias.co checkout
git config --global alias.br branch
git config --global alias.cm commit
```

◈ 15. Scenarios

- Difference between:
  - merge vs rebase
  - reset vs revert
  - fetch vs pull
- Recover deleted commits → reflog
- Move commits → cherry-pick
- Clean history → rebase -i
- Temporary save → stash

◈ Minimal Must-Know

```
git clone
git status
git add
git commit
git push
git pull
git branch
git checkout / switch
git merge
git rebase
git stash
git reset
git revert
git log
```